Leakage-Governed Medical Insurance Claim Fraud Detection Using Classical Ensembles, Tabular Deep Learning, and Dual-Path Agentic Triage

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***Abstract*—Medical insurance fraud screening suffers when classifiers exploit upstream leakage, rely on single-feature shortcuts, or lack audit transparency. Across 4,500 medical claims (6.00% fraud prevalence), we evaluate a unified platform uniting five Classical ML learners, five PyTorch Deep Learning models (MLP, ResidualMLP, CNN1D, TabTransformer, and an unsupervised Autoencoder), and a Gemini 2.0 Flash Agentic/XAI layer. Auditing the workbook uncovered two structural traps: an upstream Cluster column concentrating 98.89% of frauds (267/270) in Cluster 1, and an amount boundary separating all frauds at ClaimAmount >= $7,465.07. Across a 32-run ablation matrix (Baseline, Cluster-free Intake, No-Amount, and Temporal Intake >= 2024-03-11), tree ensembles show decisive robustness: without Cluster, Random Forest achieves 1.0000 test PR-AUC and 0.9873 F1 (0 FP, bootstrap CI [0.9610, 1.0000]), holding 1.0000 F1 chronologically while linear and neural models drop 6.6–10.7 PR-AUC points. Removing ClaimAmount collapses all models to 0.1869–0.2404 PR-AUC. On the 29-feature full-stack contract, TabTransformer leads neural models (0.9911 PR-AUC) while the unsupervised Autoencoder fails (0.0523 PR-AUC, 0.0299 F1). Exported browser weights match Python reference execution within 4.42e-7 max error across 10,800 scores with zero flips.**

Index Terms—Healthcare Fraud Detection, Data Leakage, Random Forest, TabTransformer, Precision-Recall AUC, Agentic AI, Explainable AI.

I. INTRODUCTION

Automated healthcare claim triage operates under severe class skew (often 1–10% positive rate) where black-box scalar predictions cannot satisfy Special Investigation Unit (SIU) compliance or generalize safely across hospital workflows [11, 13]. Prior evaluations frequently report optimistic random-split accuracy while overlooking upstream proxy leakage [17] and shortcut features [18]. We present a dual-track engineering and benchmark platform comparing Classical ML (Approach 1), five PyTorch neural paradigms (Approach 2), and Agentic AI + Gemini Co-Pilot XAI (Approach 3) across both a live React/Node/FastAPI enterprise system (Track A) and a browser-native research lab (Track B).

II. RELATED WORK

du Preez et al. [13] surveyed 137 healthcare fraud studies, confirming tree ensembles (Random Forests [1], XGBoost [2], histogram gradient boosting [3]) outperform linear/SVM [4] baselines on tabular claims while calling for strict temporal splits. Johnson and Khoshgoftaar [12] studied class-weighted Medicare neural models, while Huang et al. [6] and Gorishniy et al. [7] introduced attention (TabTransformer) and residual MLPs [5] for tabular learning. Grinsztajn et al. [8] proved tree ensembles surpass neural nets under irregular boundaries on ~10K-row tables. Saito and Rehmsmeier [16] proved PR-AUC must replace ROC-AUC under imbalance; Kaufman et al. [17] and Geirhos et al. [18] formalized leakage and shortcut traps; and SHAP [19], RAG [20], and ledger auditing [14] frame explainable SIU review.

III. DATASET AUDIT & PROBLEM FORMULATION

The immutable workbook holds 4,500 claims across 19 columns spanning 731 days (2022-07-09 to 2024-07-08) with zero missing entries, zero duplicates, and 270 frauds (6.00%). Six near-unique identifiers (ClaimID, PatientID, ProviderID: 4,500; DiagnosisCode & ProcedureCode: 4,495; ProviderLocation: 3,876) are removed. Cross-tabulation (Fig. 1) revealed Cluster in {0..3} concentrates 267/270 frauds (98.89%) inside Cluster 1 (24.27% vs. ≤0.26% elsewhere), and ClaimAmount sits strictly in [7,465.07, 9,978.43] for all frauds vs. [100.12, 9,997.20] overall.

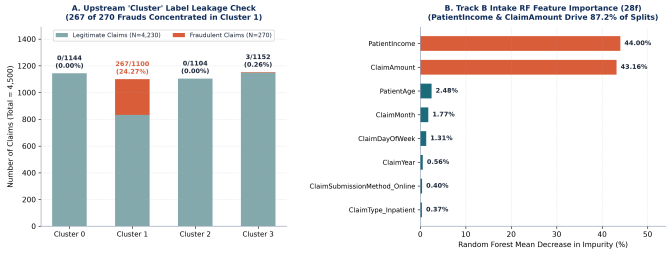


Fig. 1. (A) Cluster 1 proxy concentration (98.89% frauds); (B) Intake RF split importance led by PatientIncome (44.00%) and ClaimAmount (43.16%).

Training-only ColumnTransformer standardizes numeric inputs (plus parsed ClaimYear/Month/DayOfWeek) and one-hot encodes 7 categoricals across: (i) Track A Baseline (29f, seed 42 stratified 3,150/675/675); (ii) Track B Intake (28f, no Cluster); (iii) No-Amount (27f); and (iv) Temporal Intake (Train <= 2023-11-21 [3,152/199], Val <= 2024-03-10 [677/34], Test >= 2024-03-11 [671/37]). Supervised neural networks optimize weighted BCE with

pos_weight = N_-/N_+ ≈ 15.67, while Autoencoder optimizes legitimate-only MSE with 95th validation percentile cutoff (0.128588). Track B locks validation-F1 thresholds strictly prior to test evaluation.

IV. MULTI-PARADIGM SYSTEM ARCHITECTURE

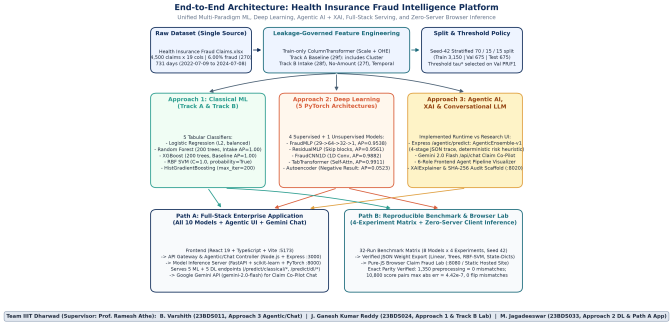


Fig. 2. Dual-path system uniting Approaches 1–3, Full-Stack Path A (React/Node/FastAPI), and Browser Parity Path B (:8080).

Path A serves 5 Classical ML and 5 PyTorch DL checkpoints (MLP, ResidualMLP, CNN1D, TabTransformer, Autoencoder) behind FastAPI (:8000) ← Express (:3000) ← React (:5173). Approach 3 delivers Express /agentic/predict (AgenticEnsemble-v1 four-stage trace), Google Gemini 2.0 Flash /api/chat co-pilot, a 6-role UI pipeline, and XAIExplainer SHA-256 logging (Fig. 3), while Path B compiles verified model weights into pure JavaScript.

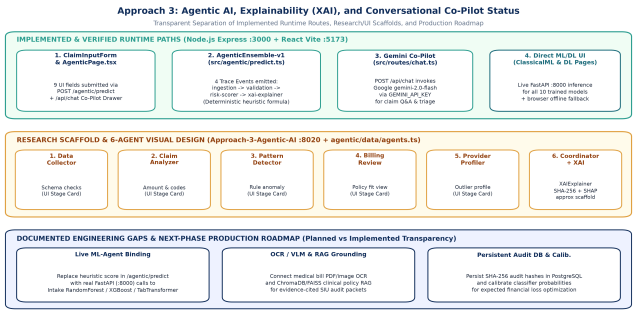


Fig. 3. Approach 3 separation of implemented Express/Gemini routes from the 6-agent UI view and roadmap.

V. EXPERIMENTAL RESULTS & VERIFICATION

A. Track A 10-Model Benchmark & Negative Autoencoder Result

Table I and Fig. 4 show Track A 29-feature test performance. XGBoost leads (100.00% PR-AUC, 98.77% F1), followed by RandomForest (99.94% PR-AUC, 98.73% F1, 0 FP). TabTransformer tops deep learners in ranking (99.11% PR-AUC, 99.94% ROC-AUC) while CNN1D leads neural F1 (90.91%, 0.77 ms). Conversely, Autoencoder collapses to 5.23% PR-AUC and 2.99% F1 because fraudulent covariates lie well inside normal ranges.

Model	Fam.	Prec	Rec	F1	ROC	PR-AUC
XGBoost	ML	97.6	100	98.8	100.0	100.00
RandomForest	ML	100	97.5	98.7	100.0	99.94
HistGradientBoost	ML	100	97.5	98.7	98.5	97.65
TabTransformer	DL	81.6	100	89.9	99.9	99.11
CNN1D	DL	83.3	100	90.9	99.9	98.82
ResidualMLP	DL	80.0	100	88.9	99.8	95.61
MLP	DL	84.8	97.5	90.7	99.7	95.38
SVM (RBF)	ML	74.5	95.0	83.5	99.6	92.52
LogisticReg.	ML	62.5	100	76.9	99.5	91.86
Autoencoder	U-DL	3.7	2.5	3.0	46.0	5.23

TABLE I. Track A Test Metrics (%) Across All 10 Models (29 Features).

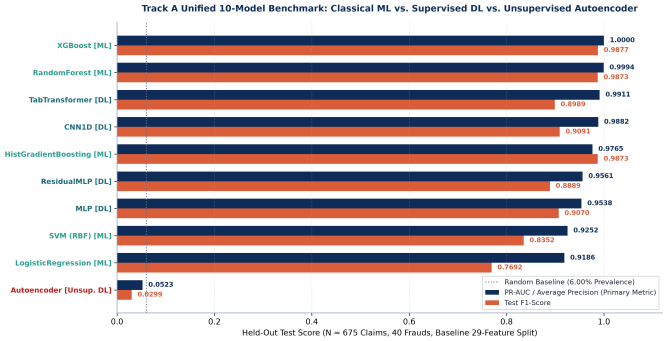


Fig. 4. Track A 10-model PR-AUC vs. F1 comparison.

B. Track B Ablation, Error Audit & Parity Ledger

Removing Cluster (Table II, Fig. 5) degrades neural/kernel models (SVM 0.9252->0.8181; ResidualMLP 0.9567->0.8765) while RandomForest achieves 1.0000 PR-AUC and 0.9873 F1 (0 FP, 1 FN) and reaches 1.0000 F1 under chronological testing. Dropping ClaimAmount plunges every classifier to 0.1869–0.2404 AP. The single Intake RF miss (Fig. 6, Row 1124) sits at the dataset minimum fraud amount ($7,465.07, score 0.2700 < tau*=0.4167; 1,000-rep bootstrap F1 CI [0.9610, 1.0000]). Browser exports achieve 0 preprocessing mismatches across 1,350 vectors and 4.42e-7 max score error across 10,800 comparisons.

Model	Base(29)	Intake(28)	NoAmt(27)	Temp(28)
RandomForest	.9994/.975	1.00/.987	.207/.364	1.00/1.00
XGBoost	1.00/.988	1.00/.988	.213/.352	1.00/1.00
GradientBoost	.977/.987	.977/.987	.240/.364	1.00/1.00
CNN1D	.988/.927	.967/.877	.229/.279	.917/.811
ResidualMLP	.957/.929	.877/.832	.225/.344	.798/.723
MLP	.954/.914	.887/.822	.230/.308	.858/.696
LogisticReg.	.919/.784	.853/.733	.232/.319	.880/.740
SVM (RBF)	.925/.897	.818/.771	.187/.304	.840/.762

TABLE II. Track B Test PR-AUC / Tuned F1 Across 4 Scenarios.

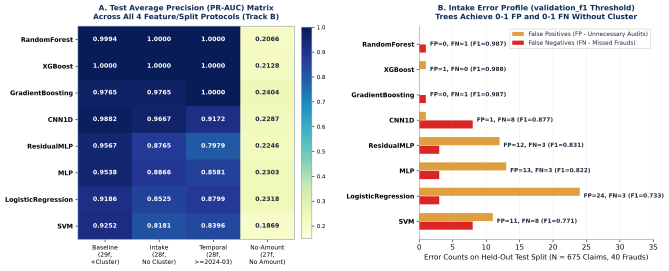


Fig. 5. Track B PR-AUC ablation matrix and Intake error counts.

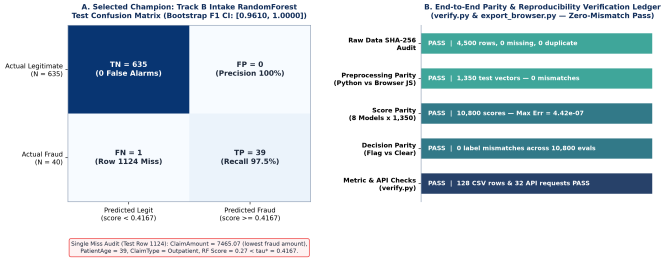


Fig. 6. Champion confusion matrix, Row 1124 audit, and parity ledger.

VI. CONCLUSION & FUTURE WORK

Empirical ablations verify that tree ensembles remain superior under leakage-free intake and chronological shifts (1.0000 PR-AUC), while TabTransformer leads supervised DL and legitimate-only Autoencoders fail under covariate overlap. Future work integrates calibrated FastAPI model calls inside /agentic/predict, clinical RAG/OCR pipelines, and low-dollar claim abuse corpora.

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